

Simone Vellei

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Grottammare (AP), Italy

PROFESSIONAL SUMMARY

Senior backend engineer with twenty years of experience building systems from the protocol layer up: custom embedded stacks in C, cloud infrastructure managing hundreds of thousands of connected IoT devices, and distributed platforms scaled to millions of concurrent resources. Active open-source author of Go frameworks for LLM pipelines and agent orchestration. Most effective at the intersection of deep technical ownership and building the small, high-ownership teams to sustain it.

EXPERIENCE

cybus.io 04/2024 - Present

Senior Backend Engineer

Full remote (English)
Cybus Connectware is the leading Industrial IoT platform for factory data integration.

- Redesigned the resource status tracking subsystem of Connectware in Go, replacing a bottlenecked component that was limiting platform scalability; the new control plane lifted the ceiling to 1 million deployable resources.
- Co-designed the migration from a synchronous RPC model to a NATS-based event-driven architecture for inter-service communication, driving key algorithmic and design decisions within a 3-person architecture team; the refactor enabled the platform to run 3 million concurrent resources, a 100x capacity improvement.
- As the sole engineer with hands-on generative AI experience at Cybus, designed and delivered a company-wide training on LLMs, embeddings, RAG, vector databases, and multi-agent architectures, driving team-wide adoption of AI-native development practices.
- Authored the architecture decision record and led a small engineering team through the design and delivery of Connectware's monitoring feature, from technical approach through rollout.
- Implemented an MCP server exposing Connectware's capabilities to AI agents, enabling agent-driven interaction with the platform.

UFirst.com 11/2021 - 04/2024

Senior Backend Engineer

Full remote
UFirst is a SaaS platform for digital queue management, virtual appointments, and customer flow analytics.

- Engineered concurrent Go microservices for a queue-and-appointment platform, identifying and offloading in-application aggregation logic to MongoDB pipelines, reducing service memory footprint under high booking load.
- Led the architectural decision to adopt Amazon Chime SDK over WebRTC for video-conference support, avoiding distributed signaling complexity; the integration enabled clients to book, manage, and conduct appointments entirely within the platform, without switching to external tools, supporting services such as remote bank mortgage consultations.
- Designed and implemented the multi-service worker pipeline for ExtrAI, a private-LLM SaaS platform for enterprise process automation, in Python, applying content-aware chunking strategies per input type to maximize LLM input quality; the system processed hundreds of conference speeches across audio, video, and slide inputs, generating per-session chatbots for attendees.

Inim Electronics 01/2017 - 11/2021

Backend Engineer / Tech Lead

- Built the company's cloud engineering function from scratch: hired and grew a team of 3, defined project management processes and code quality standards, provided direct architectural mentoring, and contributed hands-on to implementation throughout.
- Re-implemented the proprietary communication protocol stack and re-engineered the TCP/IP layer of resource-constrained burglar-alarm panels to enable cloud connectivity, keeping memory and CPU footprint within tight embedded system limits.
- Architected a cloud infrastructure to connect and manage hundreds of thousands of alarm panels deployed worldwide, designing for horizontal scalability and high device-connection density, and leading an external team through implementation.
- Built a product configurator that encoded hardware compatibility rules across model and firmware-version combinations and derived country-specific order codes from those constraints, replacing a manual and error-prone selection process.
- Researched Google Assistant and Amazon Alexa integration protocols, designed the bridge to the existing proprietary Inim cloud API, and guided the external development team through implementation to ensure protocol compliance and system stability.
- Guided a firmware team through their first embedded TCP/IP implementation: researched and selected suitable networking libraries for constrained hardware, defined the integration approach, and contributed hands-on to development alongside the team.

Senior Linux embedded engineer

- Engineered a proprietary secure internet protocol for a new generation of burglar-alarm devices, enabling home automation integration and more complex system topologies.
- Delivered the company's first GNU/Linux embedded LAN gateway for burglar alarms, adding support for third-party protocols (KNX, SIA-IP, Modbus), a responsive web interface, and email notifications.
- Implemented a token-ring network stack over RS-485 that allowed up to 30 fire-detection panels to operate as a single networked system, large enough to cover a major airport.

Bentel Security

Junior Linux embedded engineer

- Built the first prototype of a GNU/Linux-based multimedia keypad for burglar alarms and authored the product specifications for the resulting product line.

OPEN SOURCE PROJECTS

LinGoose

Go framework for building AI/LLM applications

github.com/henomis/lingoose | simonevellei.com/lingoose

- Designed and built a modular, abstraction-based Go framework for LLM application development (829+ GitHub stars, 75 forks).
- Integrates with OpenAI, Anthropic, and other LLM providers; covers embeddings, vector search, prompt pipelines, and custom extensions.
- Enabled developers to selectively import only needed components, reducing dependency footprint.

Phero

Modern Go framework for multi-agent AI systems

github.com/henomis/phero | simonevellei.com/phero

- Designed and built a composable multi-agent orchestration framework with runtime agent handoffs and MCP integration (60+ GitHub stars).
- Built-in RAG with Qdrant, PostgreSQL/pgvector, and Weaviate backends; Agent-to-Agent (A2A) protocols for inter-process communication.
- Typed observability via lifecycle event tracing, and LLM middleware for cross-cutting behaviors.

full portfolio of public projects: github.com/henomis

EDUCATION

Computer Science Master Degree

Università di Bologna, 03/2005

SKILLS

Languages: Go | JavaScript | Python | C

Generative AI: LLM API integration | Embeddings | Vector databases | RAG pipelines | Multi-agent architectures | MCP | Prompt engineering

Infrastructure & Protocols: NATS | MQTT | GraphQL | Docker | Kubernetes | AWS | GCP | CI/CD

Databases: MongoDB | Redis | MySQL | PostgreSQL/pgvector | Qdrant | Weaviate Observability: Prometheus | Grafana